

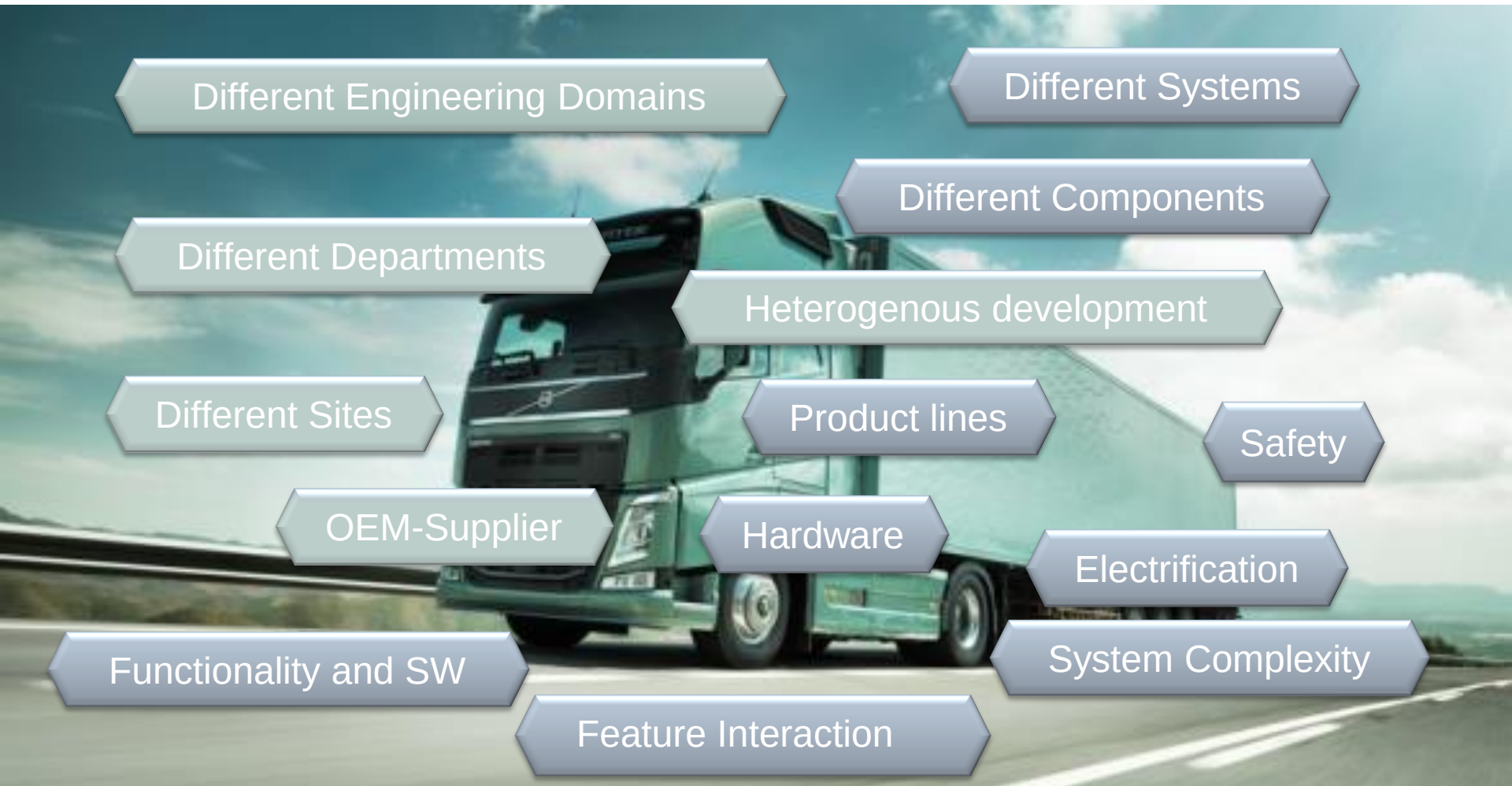
# FUSE – Synligare Seminar

II. Concepts for model exchange, views and metrics – Synligare

Henrik Lönn,  
Volvo Group, Advanced Technology and Research

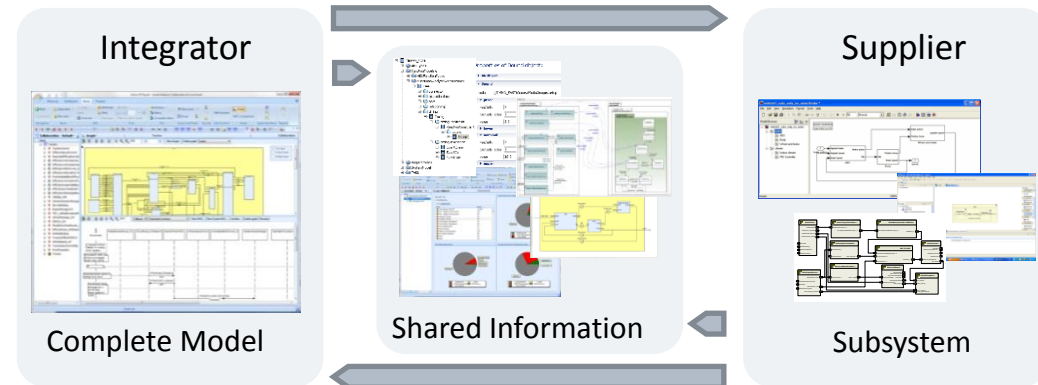
May 22 2015

# Process and Product Challenges



# Research Activities - Synligare

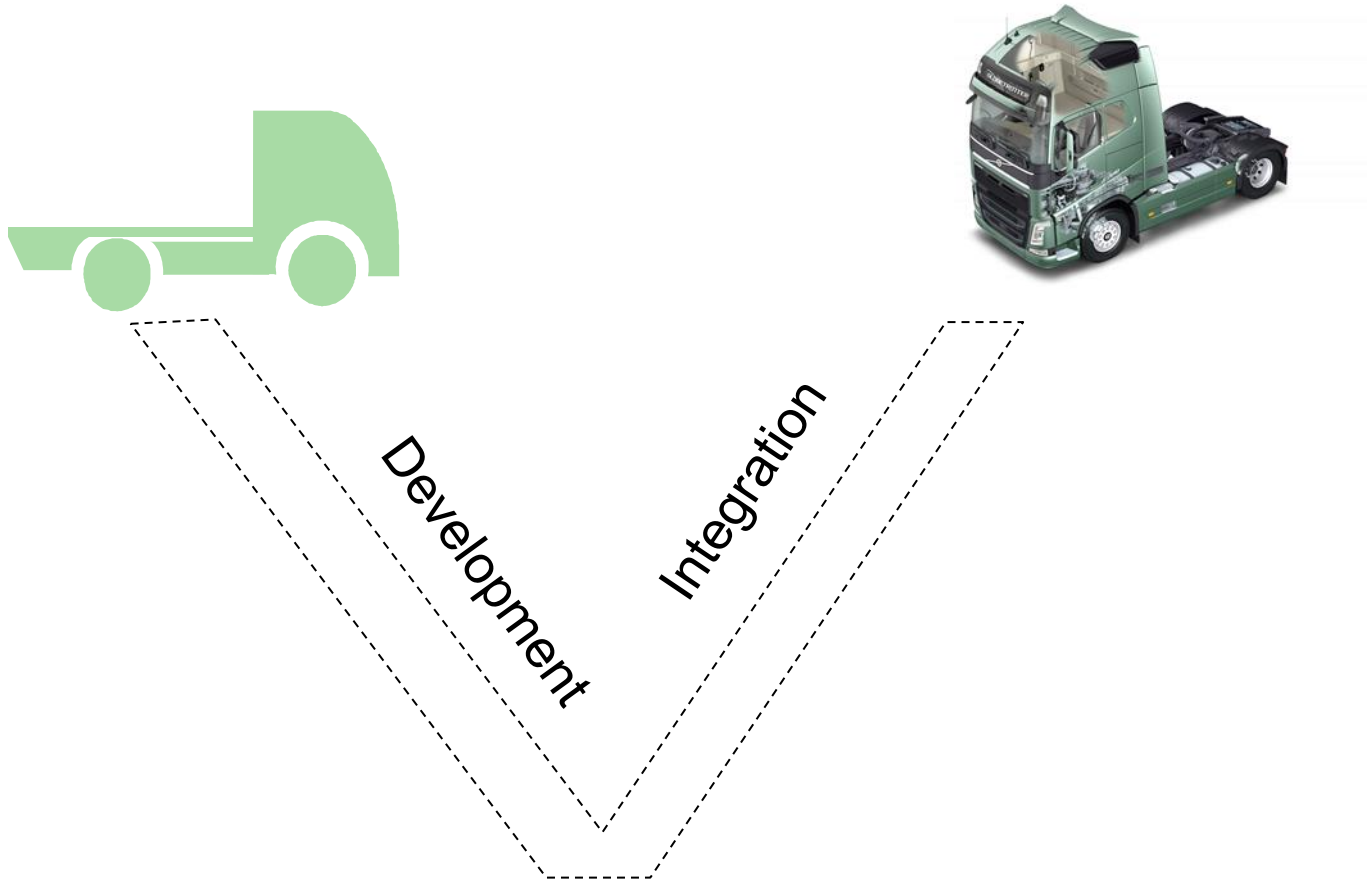
- Goal:
  - Complexity-bridging cost-effective collaborative development
- Means:
  - Improved development methodology and tools through models
- Focus:
  - Supplier – OEM collaboration
  - Requirements management
  - Views and Metrics based on models



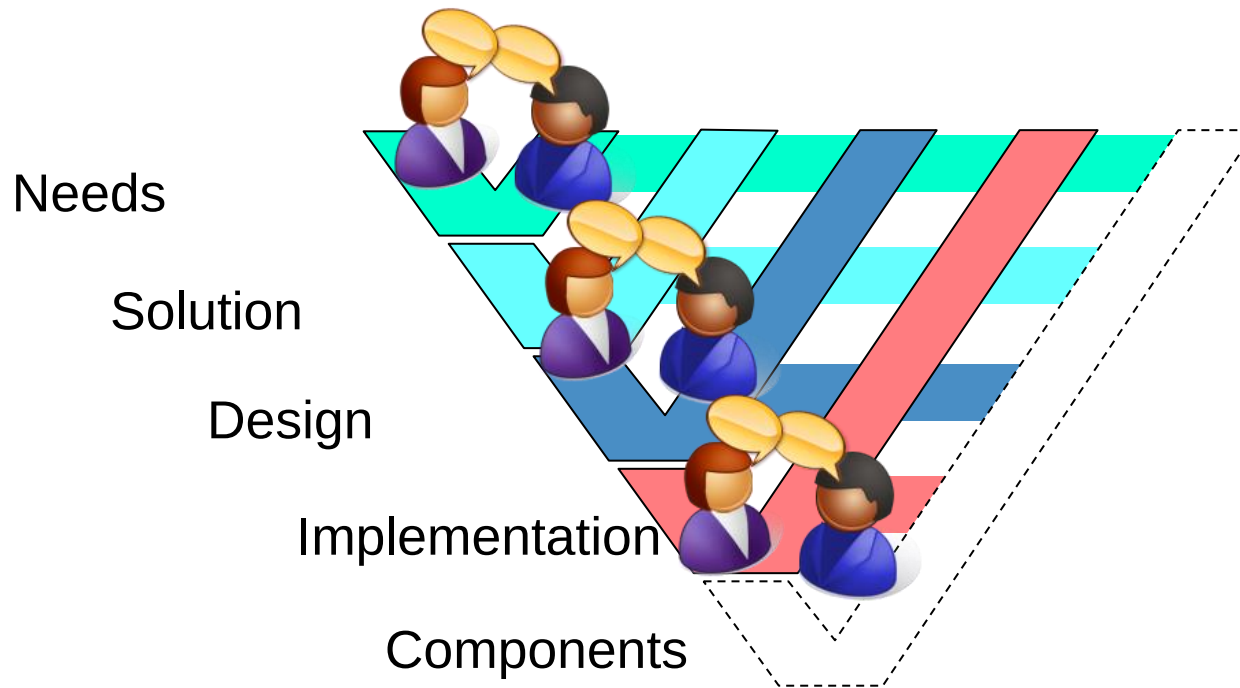
# FUSE Challenges vs. Synligare Concepts

- Model Based Approach with precise representation
- Representation of
  - System Specification
  - Safety Annotations
- Model Exchange according to agreed formats and patterns
- Views and Metrics to help understanding information

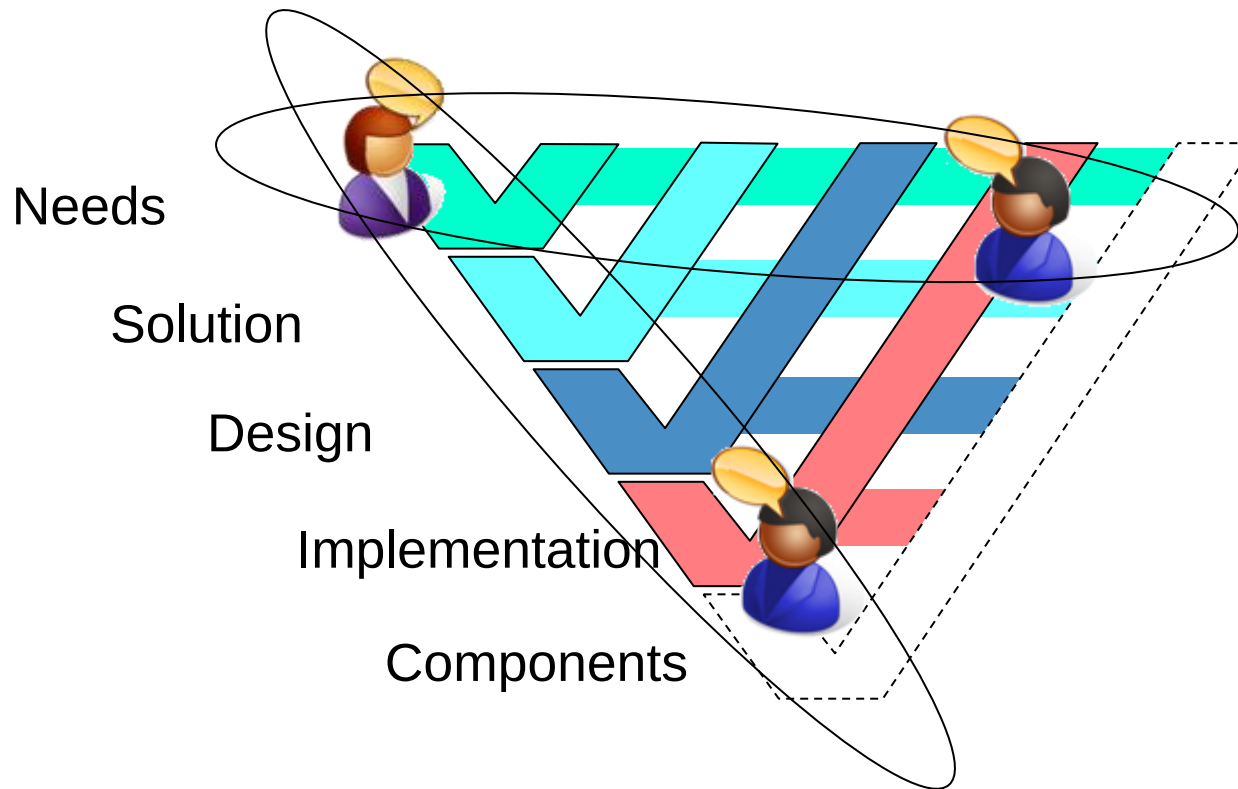
# Engineering Life Cycle



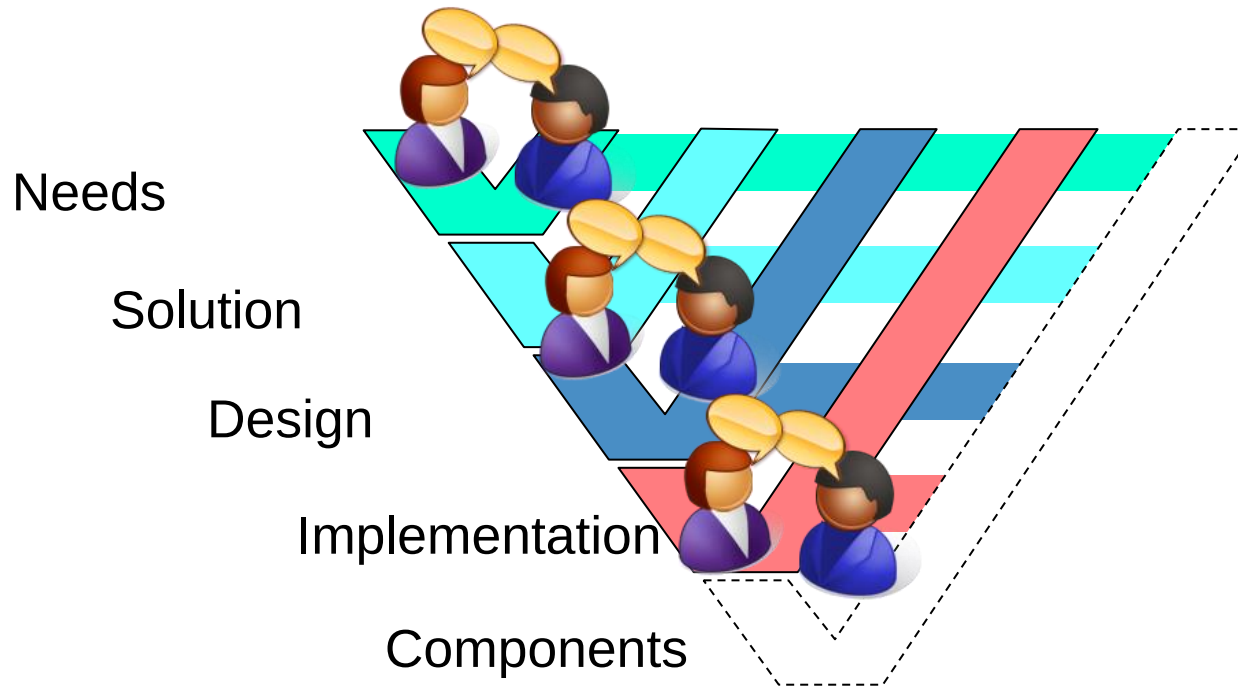
# Engineering Life Cycle – Collaboration



# Engineering Life Cycle – Iterations

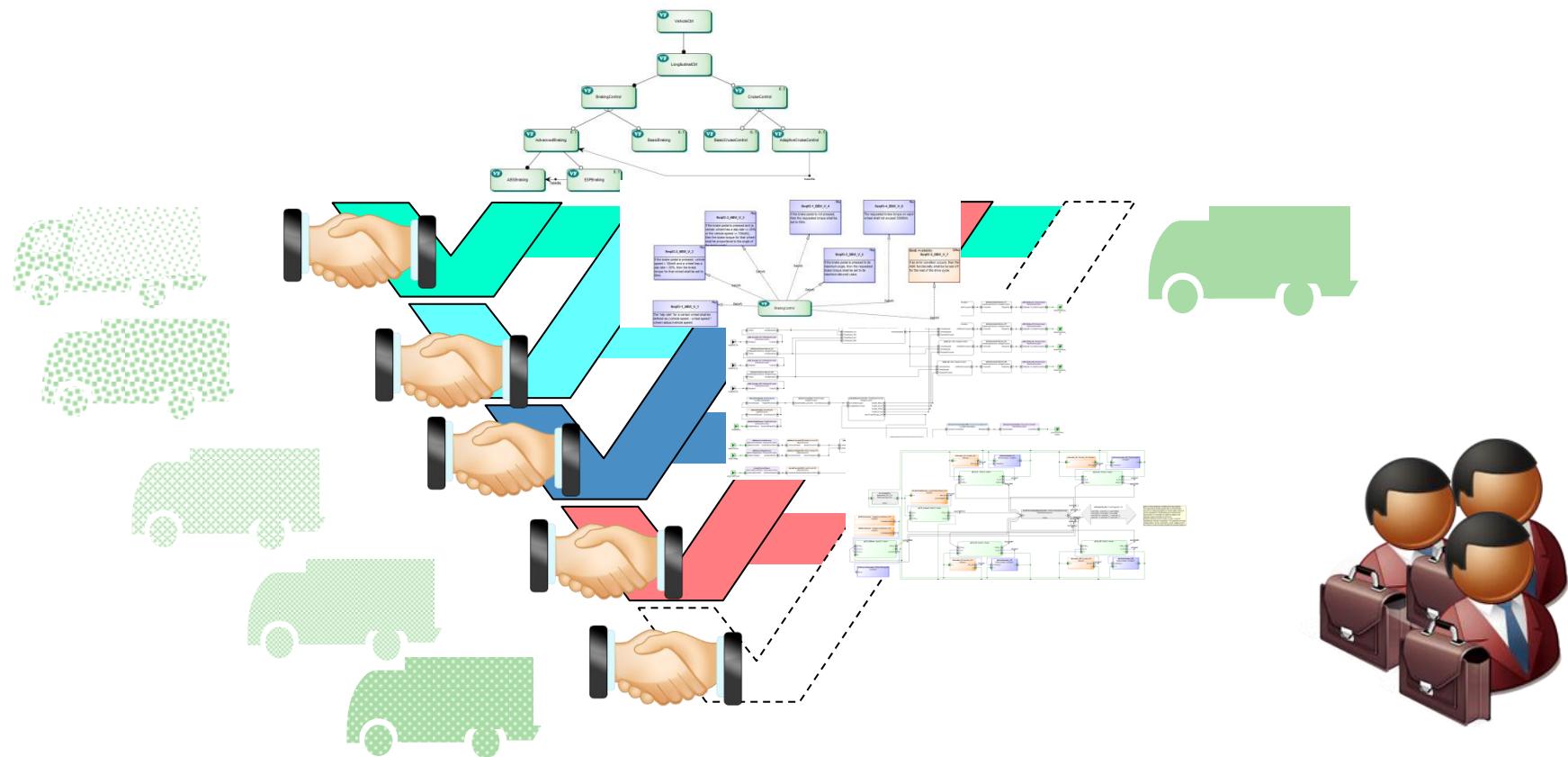


# Engineering Life Cycle – Collaboration

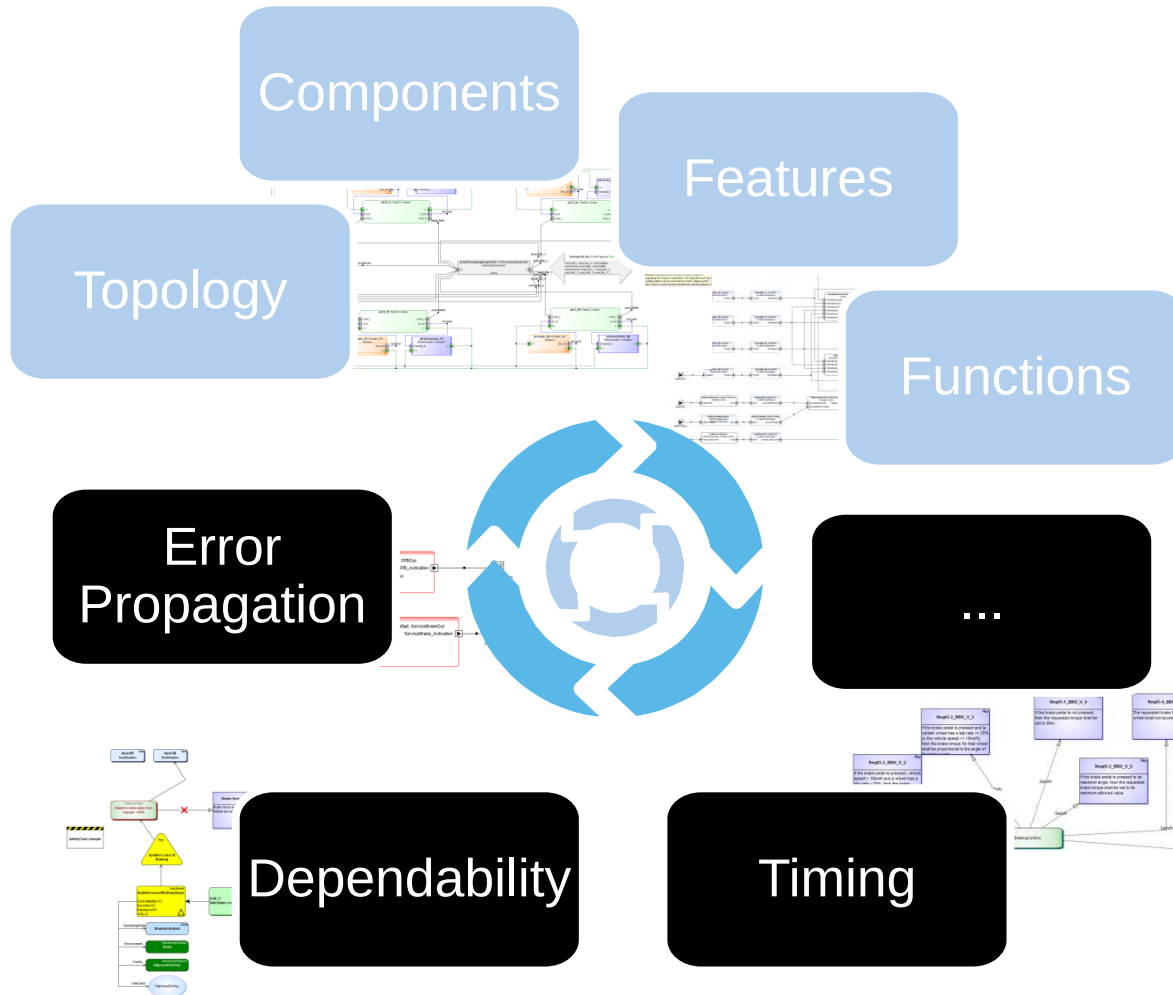




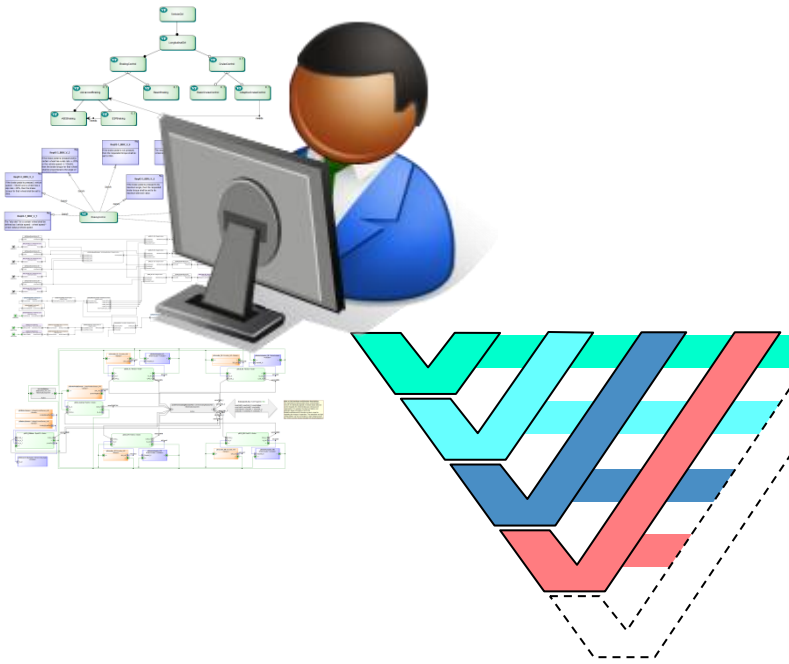
# Engineering Life Cycle – Collaboration



# Model Based Engineering



# Model Based Engineering Activities



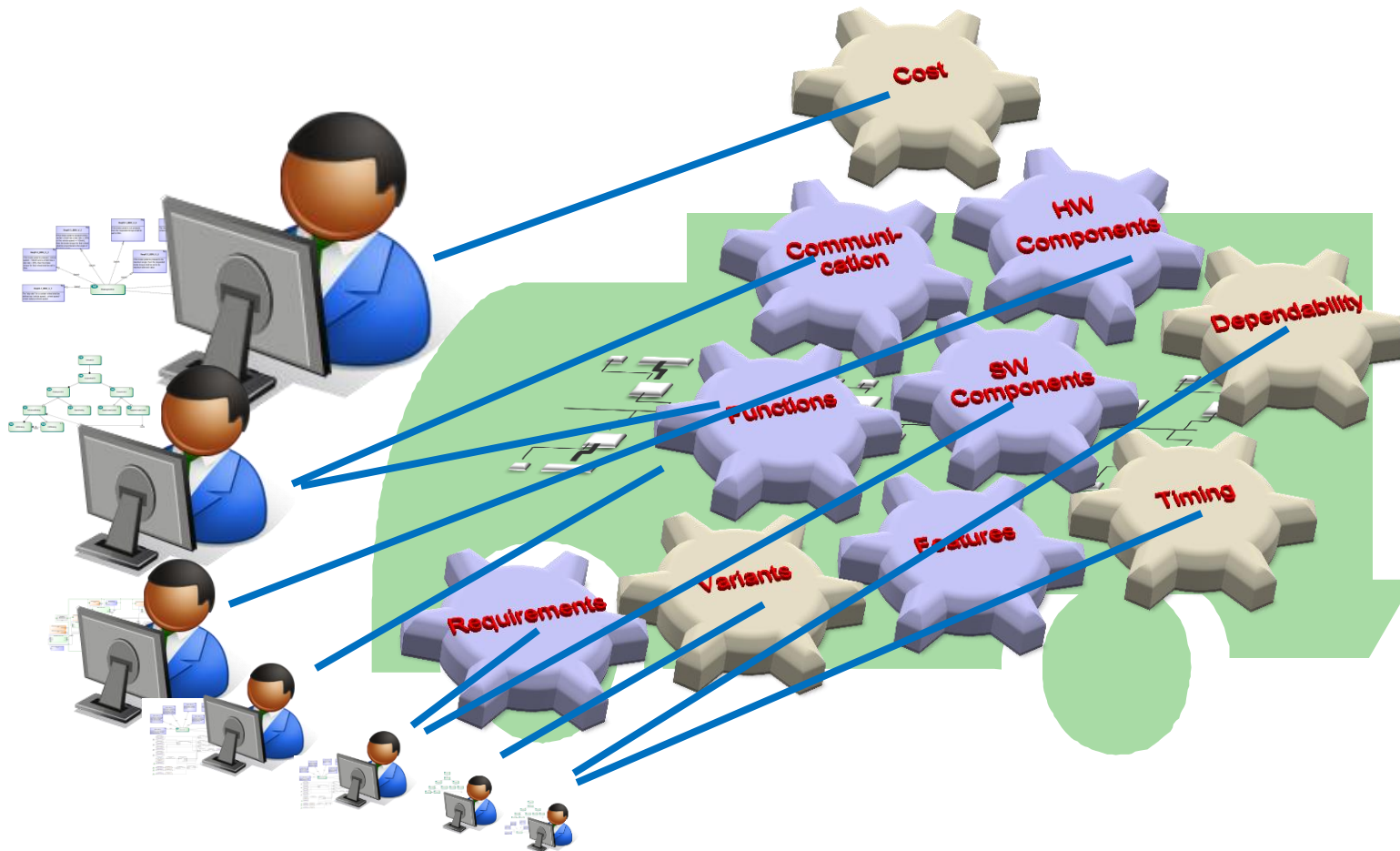
## Modelling and Synthesis

- Features
- Functions
- Topology
- Software Architecture
- Hardware Architecture
- Behavior
- Code
- ...

## Analysis and Verification

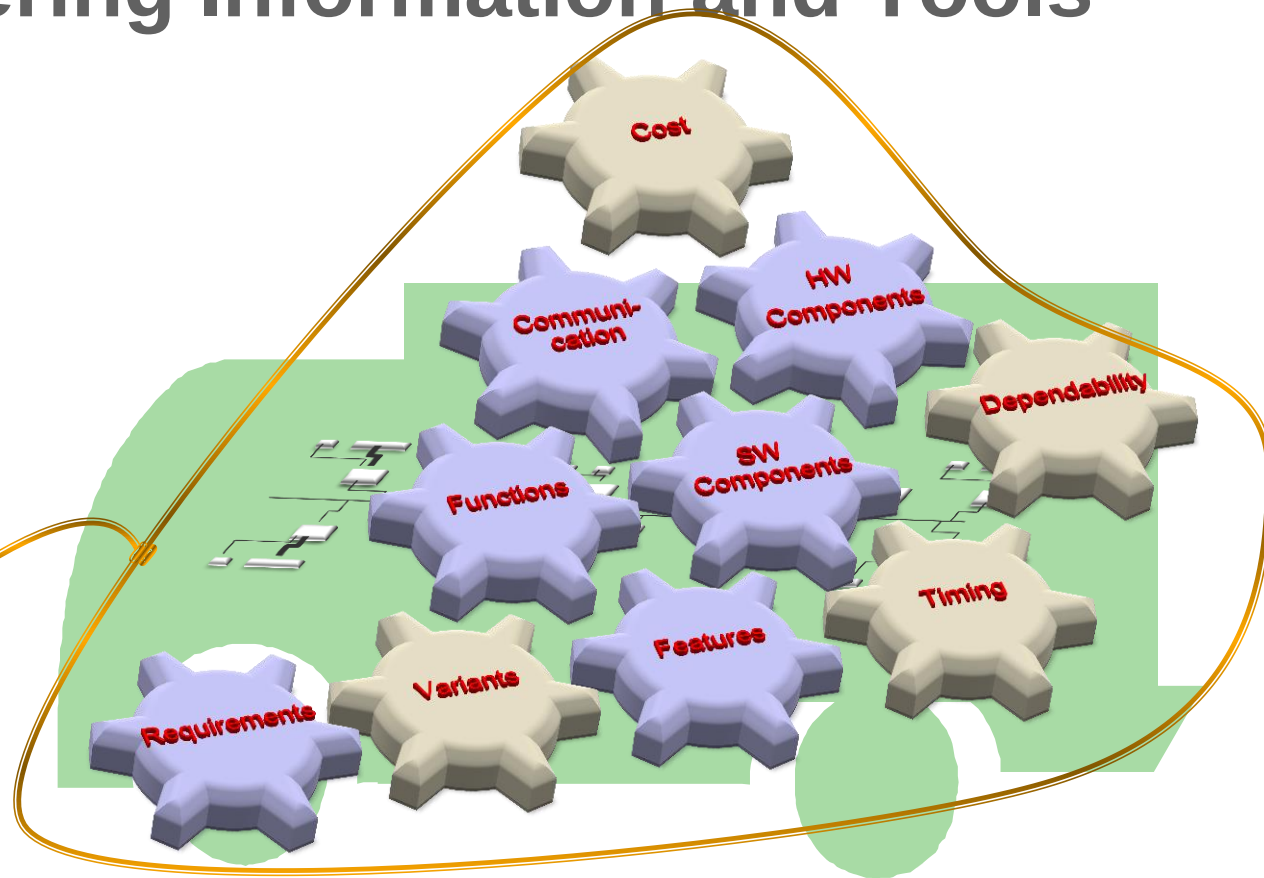
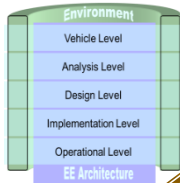
- Timing
- Dependability
- Correctness
- Energy
- Cost
- ...

# Engineering Information and Tools

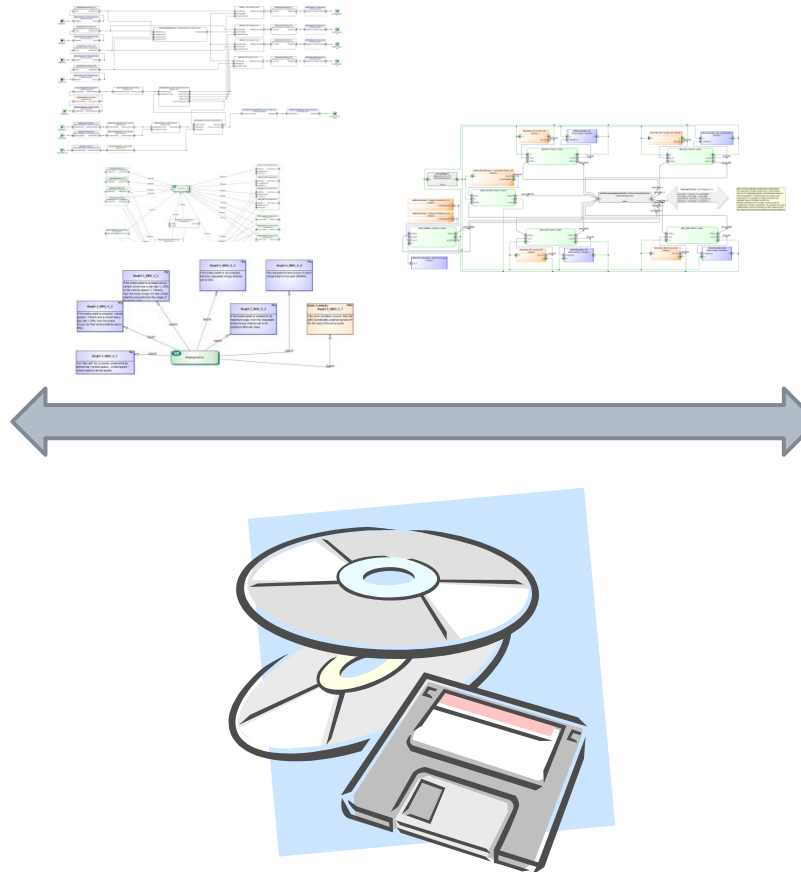


# Engineering Information and Tools

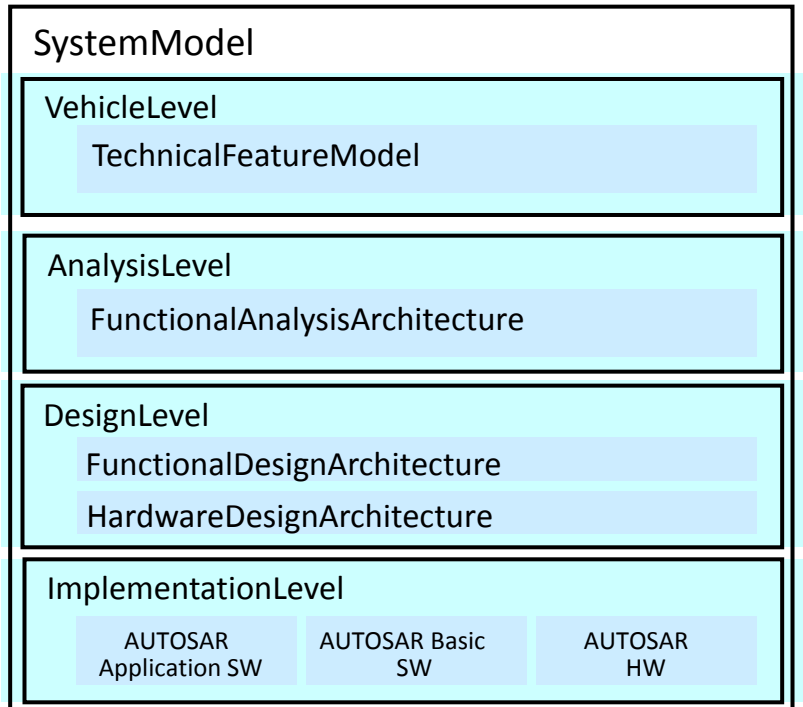
## EAST-ADL



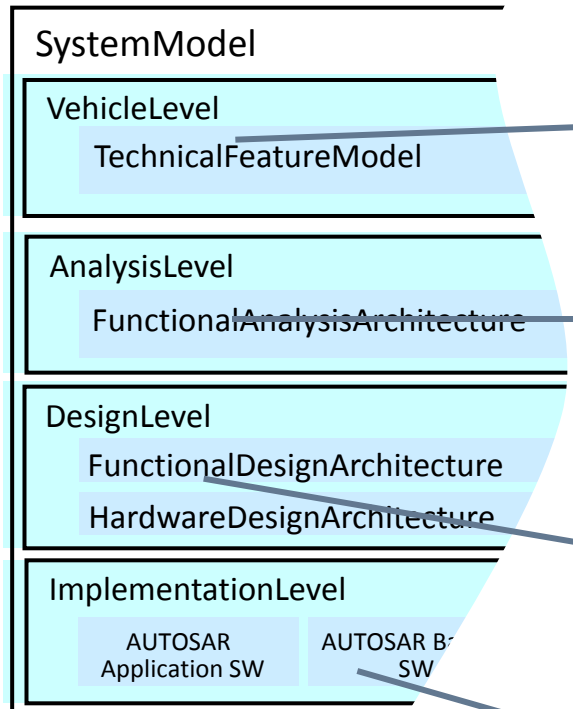
# Data Exchange



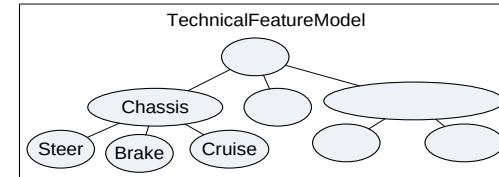
# EAST-ADL Structure



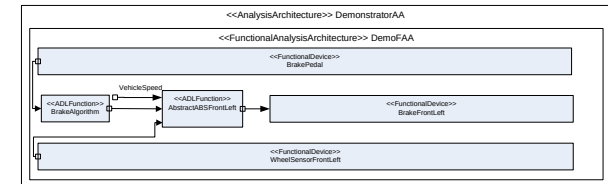
# EAST-ADL Structure



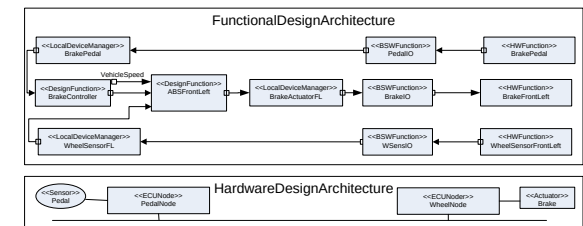
**Features  
of the vehicle**



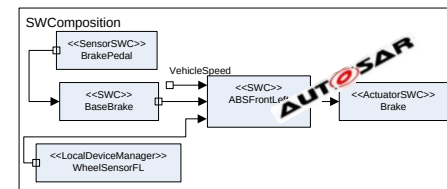
**Abstract  
functions**



**Hardware topology,  
concrete functions,  
allocation to nodes**

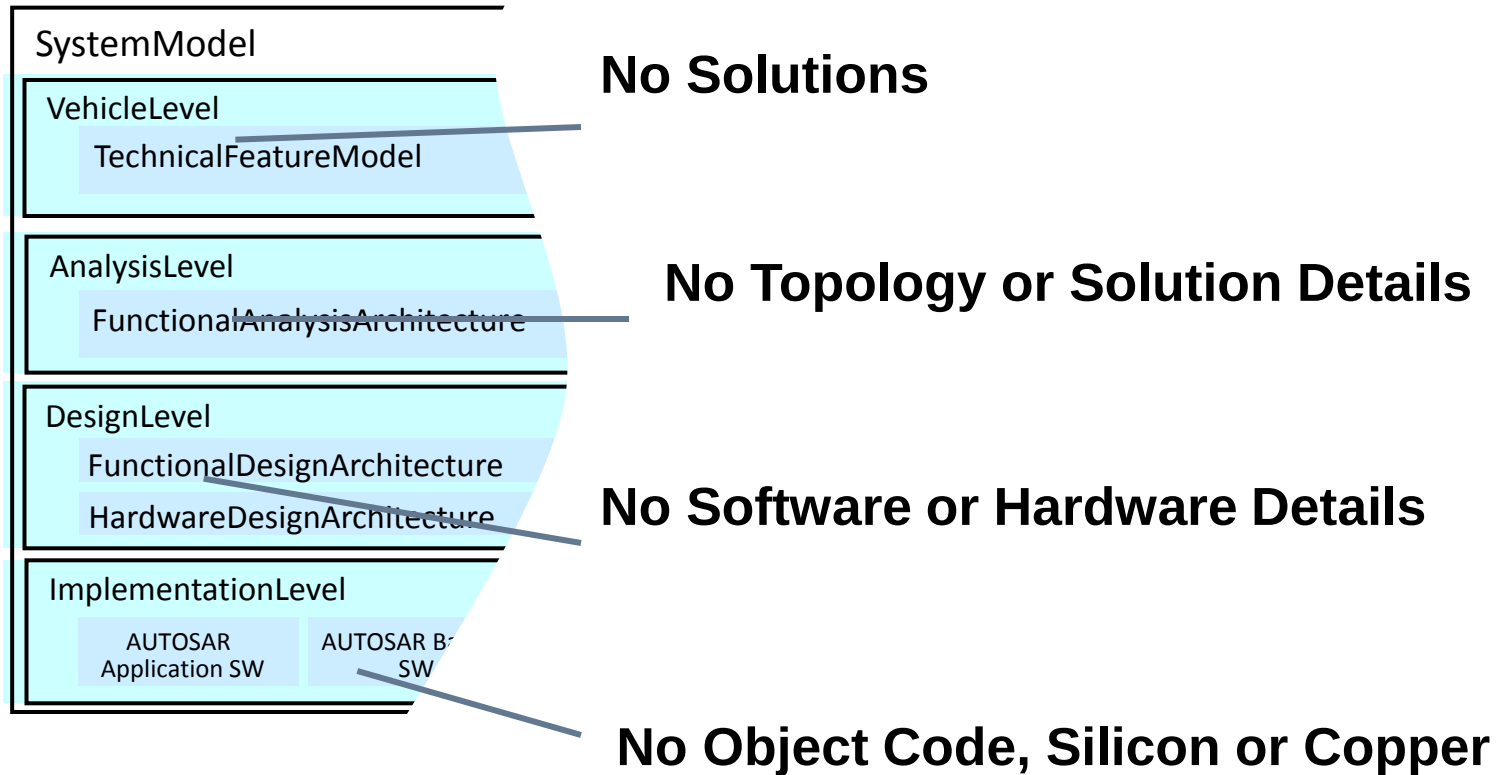


**Software Architecture  
as represented  
by AUTOSAR**

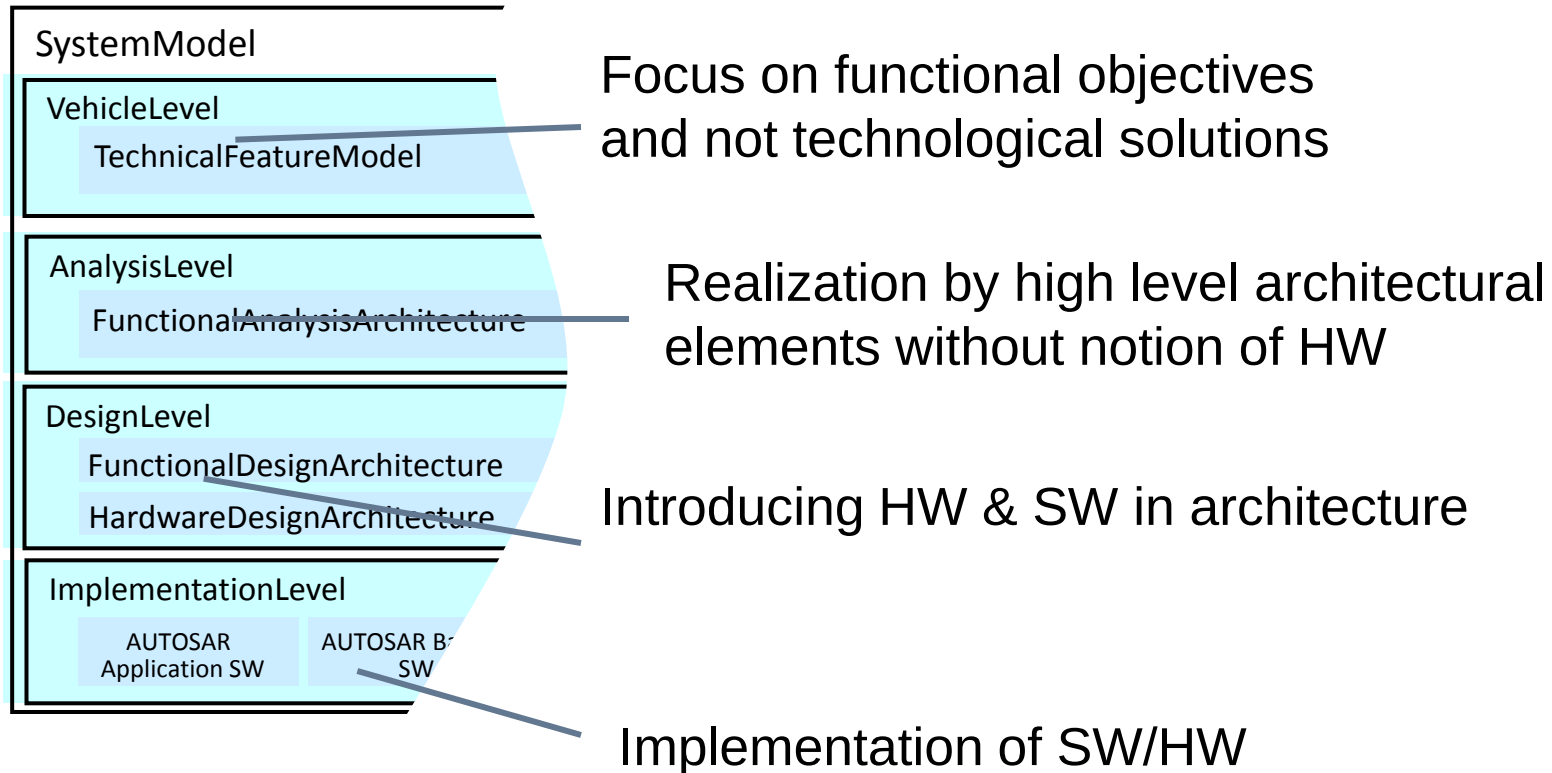




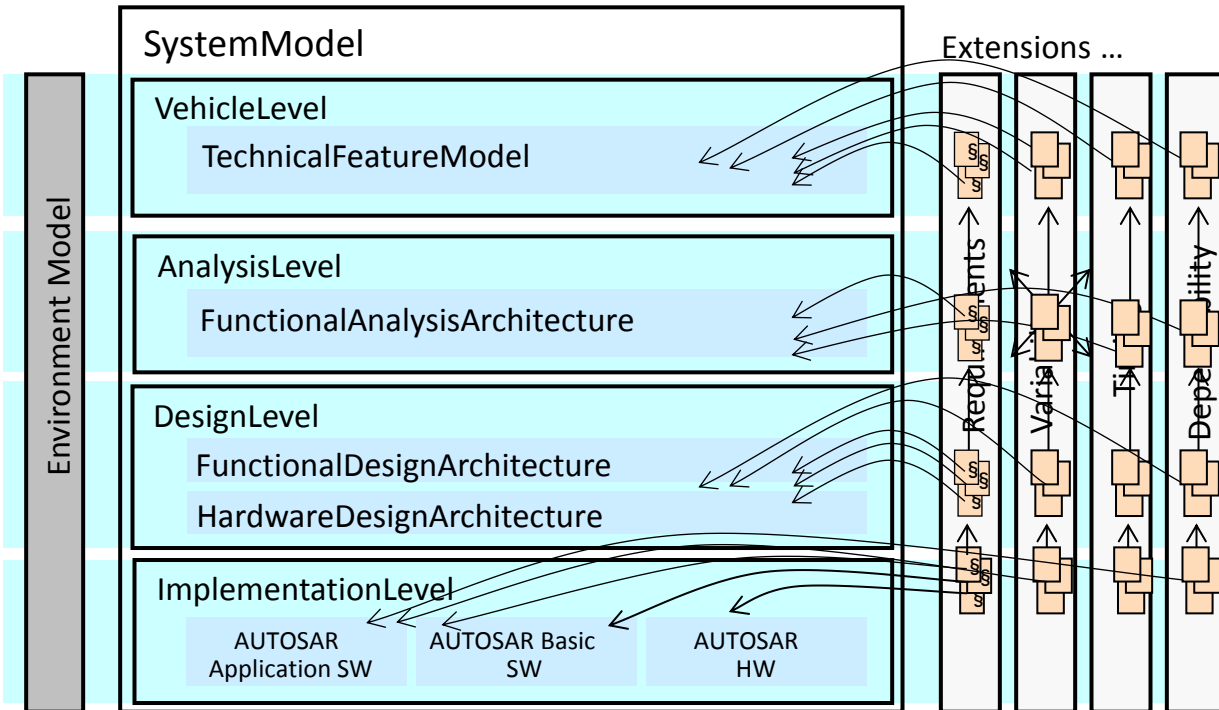
# EAST-ADL Structure



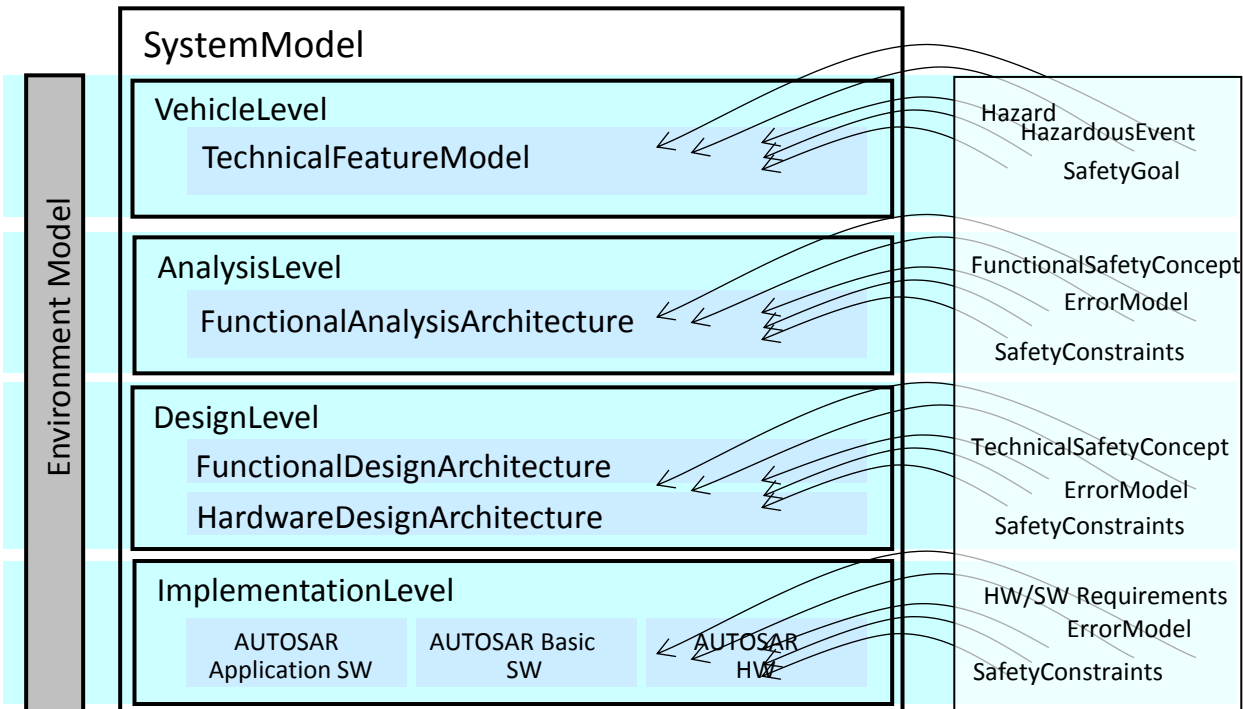
# EAST-ADL vs. ISO26262



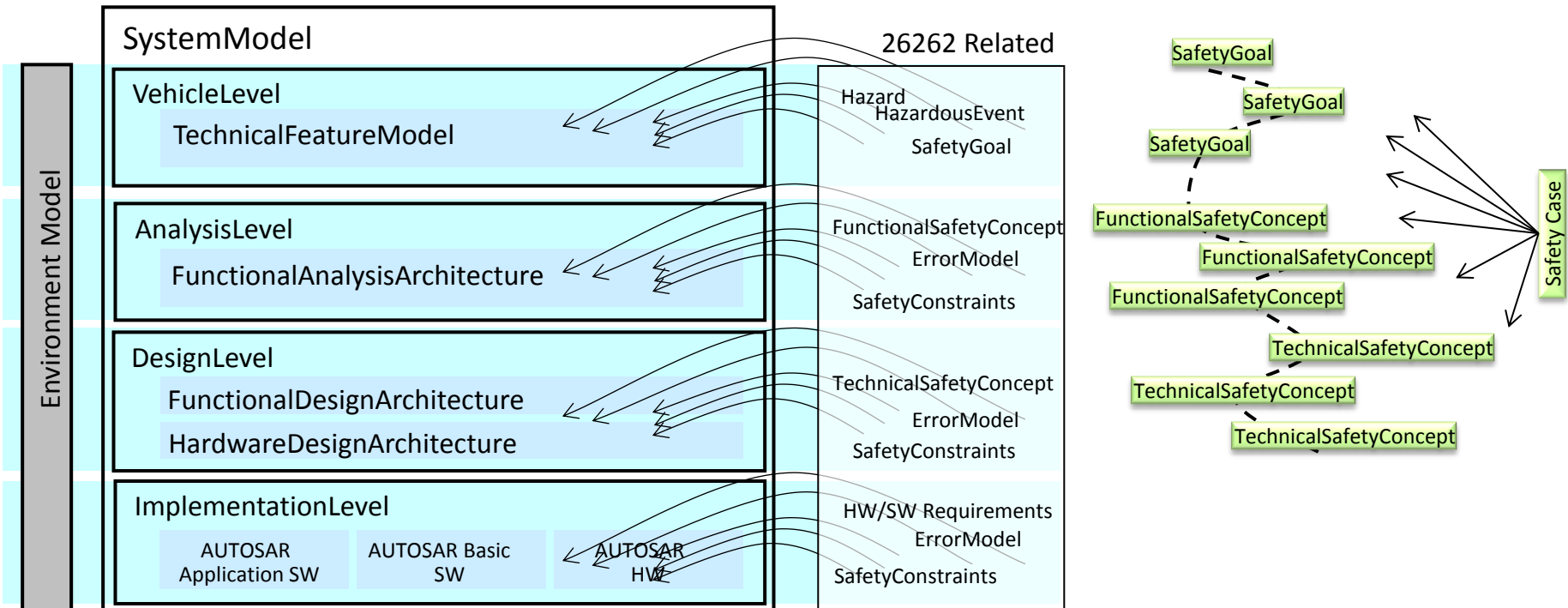
# EAST-ADL Structure

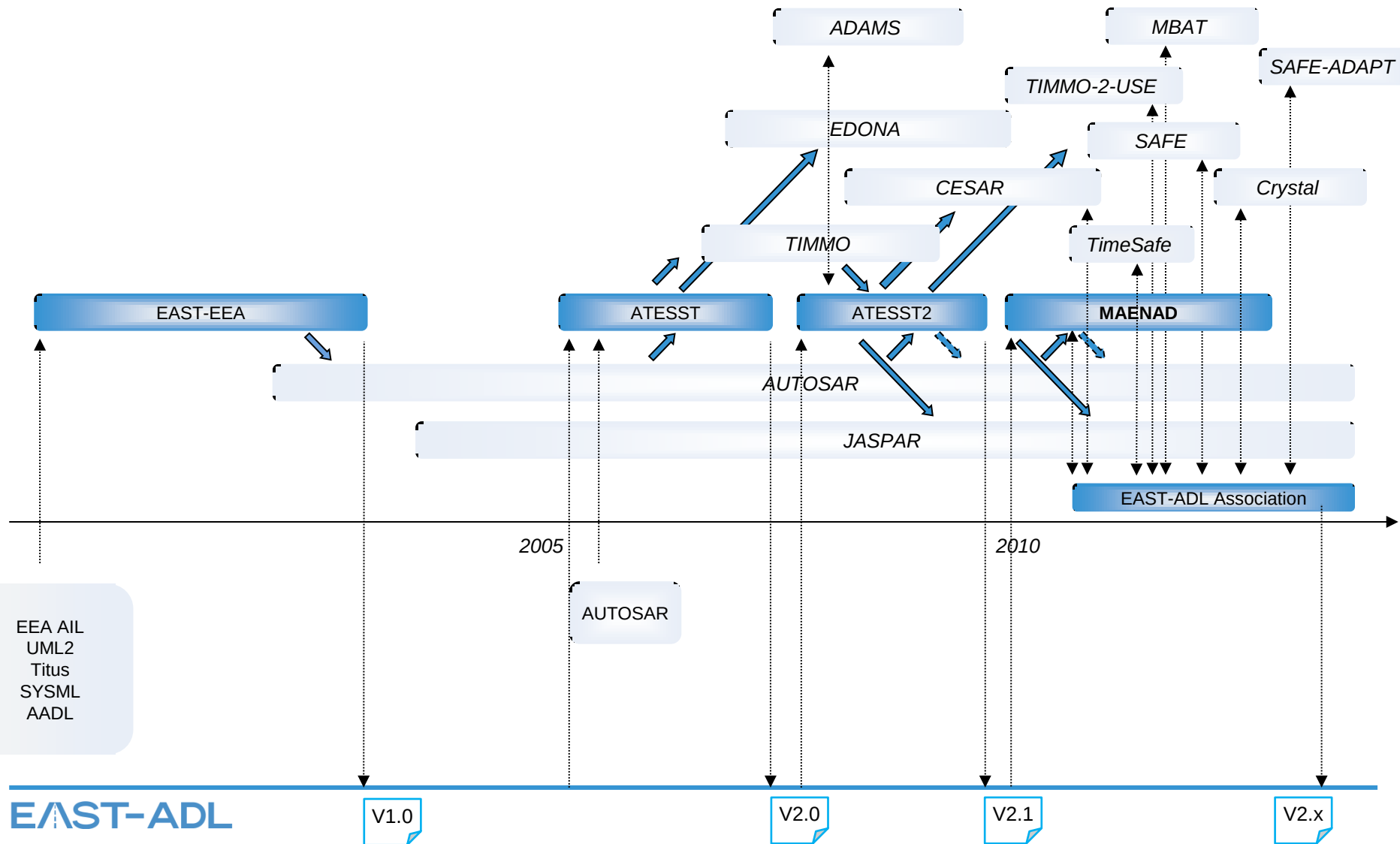


# EAST-ADL Safety Artefacts



# EAST-ADL Safety Artefacts





# Synligare Concepts

- Tools
  - SystemWeaver [systemite.se](http://systemite.se)
  - EnterpriseArchitect [sparx.com](http://sparx.com)
  - EATOP [eclipse.org](http://eclipse.org)
- Model Exchange based on EAST-ADL eaxml
- Views
  - Tables, Tracing, Tree browsing, Compare, Diagrams, Routing, ...
  - Graphical Exchange Format "sgraphml"
- Metrics
  - Product and progress metrics
  - Generic approach based on xpath



*Thank You for Your Attention!*

**EAST-ADL**  
WWW.EAST-ADL.INFO

Linked  EASTADL

**SYNLIGARE**  
**SYNLIGARE**  
**SYNLIGARE**  
**SYNLIGARE**  
www.synligare.eu